

22 Contract design

13 primary questions | Oldest to newest

2026 objectives: 3.5

11 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

Click a question below to begin.

CA1-P1 April 2007 Q5	17 marks
CA1-P1 April 2008 Q6	23 marks
CA1-P1 April 2012 Q5	19 marks
CA1-P1 September 2014 Q5	17 marks
CA1-P1 April 2016 Q1	5 marks
CA1-P1 April 2017 Q6	29 marks
CA1-P1 September 2018 Q5	17 marks
CA1-P1 September 2018 spare Q4	11 marks
CP1-P1 April 2021 Q4	8 marks
CP1-P1 April 2021 Q9	27 marks
CP1-P1 April 2022 Q2	7 marks
CP1-P1 September 2022 Q6	7 marks
CP1-P1 September 2024 Q2	5 marks

CA1-P1 | April 2007 | Q5

Current topic | 17 marks total | Original PDF p4

Related: Ch 6.

- 5** (i) State the characteristics of good design for a financial product. [5]

An insurance company is considering entering the market for long-term care contracts. These cover the costs for an elderly person of home or nursing-home care.

- (ii) Discuss the factors that will influence the design of the following areas of the new contract:

- (a) The circumstances under which the benefit is payable.
(b) The form of the benefit. [10]

- (iii) Describe with an example how two of the various design characteristics in part (i) can conflict when designing this long-term care contract. [2]

[Total 17]

CA1-P1 | April 2008 | Q6

Current topic | 23 marks total | Original PDF p4

Related: Ch 34, Ch 5.

- 6** A government is following a policy of transferring some of its operations and employees to the private sector. As part of the transfer agreements, the new employers are required to make a commitment to provide pension benefits for future service that, overall, are at least as good as those that would have accrued under the government's defined benefit scheme. The government scheme currently links increases to pensions in payment to an inflation index and provides relatively generous benefits on early or ill health retirement.
- (i) Describe the various ways of interpreting and assessing the pensions commitment. [7]
 - (ii) State any significant problems that a new employer could encounter in trying to meet its commitment. [3]
 - (iii) Outline the actions a new employer could take to make up the shortfall in benefits if they are unable to satisfy the pensions commitment fully. [5]
 - (iv) Discuss how any benefits accrued in the government scheme prior to the transfer of employment could be accommodated under the new arrangements. [8]
- [Total 23]

CA1-P1 | April 2012 | Q5

Current topic | 19 marks total | Original PDF p4

Related: Ch 18, Ch 26.

- 5 A country has a large horseracing industry. Horses generally start racing between the ages of two and five. Most horses are retired from racing when they reach the age of thirteen. A very small number of horses are retired at younger ages for breeding. Revenues from breeding fees can be very high but are very volatile.

It has been proposed that a proportion of all breeding fees should be used to provide contributions to a general fund that will be invested in order to provide benefits to look after horses in retirement.

All horses actively involved in racing would be covered by the fund.

Benefits would normally only be payable when horses retire from racing aged thirteen.

Benefits would only be available in respect of retirement before age thirteen if the horse is certified as injured by a registered vet.

No benefits would be payable on the death of the horse either before or after retirement.

- (i) Outline the demographic information that would need to be considered when determining the cost of any retirement benefits. [5]

Two possible forms of benefits have been proposed:

- (a)

Benefits for each horse will be paid out of the general, overall resources of the fund

Benefits will be calculated as a cash amount for each year of a horse's racing career to be paid as an annual income from retirement until the horse dies. This amount will be the same for all horses and will be increased each year, both before and after retirement, in some way to allow for inflation.

- (b)

Benefits for each horse will be provided from individual accounts held within the overall fund

A proportion of each year's contributions will be allocated to each horse. These contributions will form individual accounts for each horse. The value of these accounts will grow in line with investment returns earned by the general fund. On retirement, the accumulated individual accounts will be used to provide appropriate benefits.

In both cases, any proceeds would be payable to the horse's owner who would use them on behalf of the horse. Measures would be taken to ensure that any payments are actually used to provide care for horses and not taken by their owners.

CA1-P1 | April 2012 | Q5 - continued

[Current topic](#) | [19 marks total](#) | [Original PDF](#) p5

Related: Ch 18, Ch 26.

- (ii) Discuss the relative attractions of each proposal in terms of providing suitable benefits for horses retiring at age thirteen or on earlier certified retirement.

[10]

As an alternative to proposal (a), it has been suggested that instead of paying income until the horse dies, the fund will reimburse the cost of housing the horse at a “retirement farm” approved by the fund.

- (iii) Describe the additional risks that the fund may face if this alternative were adopted.

[4]

[Total 19]

CA1-P1 | September 2014 | Q5

Mixed / scope note | 17 marks total | Original PDF p3

Includes historical project planning or appraisal material not explicitly listed in 2026. The risk, modelling and commercial applications remain relevant; keep the full question together. Related: Ch 30, Legacy: project planning and appraisal techniques.

- 5** Susan was employed as a sports teacher at a school. As part of her employment package she was a member of the group insurance scheme which covered many perils from professional negligence to sickness cover. She has recently become self-employed and set up her own sports coaching business.

- (i) State two forms of insurance she may need in her new self-employed status. [1]

A newly established, well-capitalised proprietary insurance company specialises in niche markets. Its latest project is to develop a public liability insurance product for self-employed individuals. This product would pay damages if someone is injured as a result of the policyholder's work.

- (ii) Discuss aspects that the insurance company will need to consider when planning this project. [10]
- (iii) Describe two scenarios in which Susan might claim under this product. [2]
- (iv) Set out the information about Susan's work that the insurance company will require if she applies for this product. [4]

[Total 17]

CA1-P1 | April 2016 | Q1

[Current topic](#) | [5 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 1** A general insurance company is planning to offer a new contents insurance product for homeowners. They will offer a guarantee that the premiums for contents insurance will only rise in line with inflation for five years. This guarantee will be offered for an agreed sum assured.

Discuss the issues around managing this guarantee. [5]

CA1-P1 | April 2017 | Q6

Current topic | 29 marks total | Original PDF p4

Related: Ch 18, Ch 29, Ch 30.

- 6** A large financial institution currently has no operations in the insurance sector.

It has decided to expand into this market by bringing out a new whole life product. This product will provide a fixed cash sum intended to cover the policyholder's funeral expenses and other related costs.

The product will be marketed as simple and low cost for policyholders.

The product will be advertised in newspapers and on television by an appropriate celebrity.

There is a guarantee that every applicant between the ages of 50 and 75 will be accepted subject to only minimal underwriting.

- (i) Set out what the minimal underwriting may consist of. [3]
- (ii) Outline the practical problems that may arise due to the lack of data held by the institution. [3]
- (iii) Describe the commercial implications of these data problems and of the other features of the policy. [6]
- (iv) State the criteria for selecting the celebrity advertising the product. [2]
- (v) List incentives that could be offered to attract policyholders. [2]
- (vi) Outline why some claims might be excluded under the policy. [2]

The institution is worried that development of the new line of business may be made difficult due to a lack of relevant expertise and management systems.

They are therefore considering two other options for introducing the new product and hence entering the insurance market:

- Purchase an existing life insurance company; or
 - Enter into a joint venture. That will create a new company jointly owned by the institution and an existing insurance company to write the new product.
- (vii) Explain how each of these options could overcome the data, expertise and systems problems that would otherwise be faced by the institution, and any potential problems to be considered before proceeding with either option. [11]
[Total 29]

CA1-P1 | September 2018 | Q5

Current topic | 17 marks total | Original PDF p3

September 2018 main paper, recovered from the supplied archive; the extracted folder had contained the spare paper under this date. Related: Ch 26.

5 Two companies in the same industry have approached an actuary to advise them on the benefits packages they offer their respective employees.

- PXU has been established for over 25 years and has around 1,000 employees based in five locations around the country.
- Unicorn was set up six months ago and has ten employees, based in a single location in the same country.

(i) Outline how the issues to be considered, in designing the benefits packages, may differ between the two companies. [4]

The actuary proposes to both companies that they offer a flexible benefits system. Under this arrangement, employees are offered the option at the start of each year to revise the level of an existing benefit or to choose different benefits, which the employees “buys” either by reducing their pay or by giving up part of their existing benefits.

(ii) Describe how a flexible benefit approach can help meet the different needs of employees. [2]

As part of its new flexible benefits programme, PXU decides to partner with an insurance company, GGG, to offer life insurance cover to its employees.

(iii) Set out the main risks involved in this arrangement for:

- (a) PXU.
- (b) The employees of PXU.
- (c) GGG.

[6]

Unicorn proposes to award its employees shares in the company at the end of each year, based on the individual employee’s performance over the year.

(iv) Discuss the practical issues that Unicorn needs to consider in implementing this proposal. [5]

[Total 17]

CA1-P1 | September 2018 spare | Q4

[Current topic](#) | 11 marks total | [Original PDF](#) p2

2018 spare paper (X18), distinct from the September main sitting. Original paper and matching spare examiner report restored from the supplied archive.

- 4** A company is designing a remuneration package for its employees that will include linking benefits with corporate objectives.
- (i) Describe how the company could design a remuneration package which can link benefits with corporate objectives. [6]
- A company employs investment managers to manage long term unit linked investment bonds.
- (ii) Outline the considerations for designing a remuneration package for these employees. [5]
- [Total 11]

CP1-P1 | April 2021 | Q4

Current topic | 8 marks total | Original PDF p3

Related: Ch 37.

- 4** A life insurance company currently offers its with-profits policyholders the option of surrendering their policy early. On surrender, the policyholder is paid a surrender value, the amount of which is at the discretion of the company. Historically, surrender values paid in the early years of a policy have been significantly lower than the expected value of the policy at maturity, even after adjusting for future premiums and investment returns.

- (i) Outline the reasons why the life insurance company would take the approach outlined above to determine surrender values. [2]
- (ii) Suggest possible reasons why a policyholder would want to surrender the policy in the early years. [2]

Following complaints from surrendering policyholders, the insurance company plans to review its approach to determining surrender values.

- (iii) Suggest possible factors the insurance company should consider in its review. [4]
- [Total 8]

CP1-P1 | April 2021 | Q9

Current topic | 27 marks total | Original PDF p5

Related: Ch 20, Ch 21, Ch 15, Ch 38.

- 9** A football club has a defined benefit pension scheme for its employees. It has three different categories of employee who work for the club.
- Players – generally paid between £500,000 and £1,000,000 p.a. At this particular club, they are very loyal and play for the club between the ages of 18 and 35, at which point they retire.
 - One manager – the club has employed five managers over the last 60 years; the current manager is paid £250,000 p.a.
 - Staff – the remaining 100 staff members have worked for the club for an average of 30 years, with an average salary of £40,000 p.a.
- (i) Outline the possible issues the football club would have considered in designing the benefits for each category of employee. [7]
 - (ii) Explain why different mortality assumptions may need to be used for each category of employee. [5]
 - (iii) Discuss how the scheme's managers could control the different expenses associated with running the pension scheme. [5]
 - (iv) Suggest, with reasons, an appropriate investment strategy for the scheme. [7]
 - (v) Explain the importance of monitoring the scheme's experience against the scheme's funding assumptions. [3]
- [Total 27]

CP1-P1 | April 2022 | Q2

Current topic | 7 marks total | Original PDF p2

Related: Ch 30.

- 2** An insurance company sells single premium unit-linked investment bonds. The bonds can be surrendered at any time by the policyholder for the full value of the units at the date of surrender.

The company is to introduce a new version of the bond with an attaching guarantee. Under the terms of the guarantee, the surrender value on the 5th and 10th policy anniversaries will be subject to a minimum of the single premium paid.

- (i) Outline how the company's administration systems will be affected by the introduction of the new version of the bond with a guarantee. [2]
- (ii) Outline how the company could manage the possible financial costs of the guarantee. [5]

[Total 7]

CP1-P1 | September 2022 | Q6

Current topic | 7 marks total | Original PDF p3

Related: Ch 10.

- 6** An insurance company uses financial derivatives to manage its exposure to options and guarantees across a range of financial products it has sold to investors. The range of financial products includes ones that pay benefits on death, on surrender and on maturity.
- (i) Outline the options and guarantees that the insurance company could offer at death, surrender and maturity. [3]
 - (ii) Explain why the insurance company may use financial derivatives to manage risk in these products. [4]
- [Total 7]

CP1-P1 | September 2024 | Q2

Current topic | 5 marks total | Original PDF p2

Related: Ch 4.

- 2** A financial provider is developing a new pension product that will be targeted at individuals who have recently graduated from university.
- Describe the main factors the provider should consider when developing this new product. [5]